

Composition Operators on Sobolev Spaces in Metric Measure Spaces

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Motivating question

The simplest change-of-variables problem: I

Let AC^p , $1 \leq p < \infty$, be the seminormed space of absolutely continuous functions u such that $u' \in L^p$, $\|u\|_{AC^p} = \|u'\|_{L^p}$. Let $\varphi: [a, b] \rightarrow [c, d]$ be a homeomorphism.

We consider the change-of-variables problem on AC^p : Suppose that

$$\varphi^* u := u \circ \varphi \in AC^p[a, b] \quad \text{for every } u \in AC^p[c, d],$$

and there exists a constant $K > 0$ such that the inequality

$$\|\varphi^* u\|_{AC^p[a, b]} \leq K \|u\|_{AC^p[c, d]} \quad \text{holds for every } u \in AC^p.$$

We say that φ^* is a **composition operator**.

The simplest change-of-variables problem: II

Question

What geometric restrictions on φ follow from the boundedness of the change-of-variables operator? Are they sufficient?

Answer

For $p > 1$, homeomorphism φ must be Lipschitz. Determine the $\|\varphi^*\|$.

What is the corresponding result for $p = 1$? Determine the $\|\varphi^*\|$.

Preliminaries



Preliminaries: I

- A triple (X, d, μ) is a **metric measure space**, if (X, d) is a nonempty separable metric space, μ is a nontrivial regular Borel measure.
We also assume that μ is finite on bounded subsets.

Examples

- Euclidean spaces $(\mathbb{R}^n, d_{eu}, \mathcal{L}^n)$,
- Riemannian manifolds $(M^n, d_{riem}, \text{vol}_g)$,
- stratified Lie groups (also known as Carnot groups) or subRiemannian manifolds,
- open domains in the above spaces,
- and many other examples.

Preliminaries: II

- The Dirichlet space $D^{1,p}(X)$, $1 \leq p < \infty$, consists of measurable functions $u: X \rightarrow \mathbb{R}$ such that the inequality

$$|u(x) - u(y)| \leq \int_{\gamma} g \, ds$$

holds for all curves γ with endpoints $x, y \in X$. Here $g \in L^p(X)$ is a nonnegative Borel function independent of the curve γ and of the points x, y .

- Let $u \in D^{1,p}(X)$. There exists a L^p -minimal upper gradient. We denote it by $|\nabla u|_p$. The smallest upper gradient $|\nabla u|_p$ may depend on p .

For notational simplicity, we write $|\nabla u|$ instead of $|\nabla u|_p$.

Preliminaries: III

- We say that a measurable mapping $\varphi: X \rightarrow Y$ belongs to the Reshetnyak class $D^{1,p}(X; Y)$, $1 \leq p < \infty$, if
 1. $u \circ \varphi \in D^{1,p}(X)$, for every $u \in \text{Lip}(Y)$ with $\text{Lip}(u; Y) \leq 1$, and
 2. there exists an L^p -envelope w such that

$$|\nabla(u \circ \varphi)| \leq w \quad \text{a.e.}$$

The smallest envelope w is denoted by $|D\varphi|$.

$$X \xrightarrow{\varphi} Y$$

$$D^{1,p}(X) \xleftarrow{\varphi^*} \text{Lip}(Y)$$

Main results

Characterization of composition operators

Theorem (Sboev, 2025 ($p = q$); Vodopyanov, Sboev, 2025, 2026 ($q < p$))

Let (X, d, μ) and (Y, ρ, ν) be metric measure spaces, and let $\varphi: X \rightarrow Y$ be a homeomorphism. Let $1 \leq q \leq p < \infty$.

The composition operator $\varphi^*: D^{1,p}(Y) \cap \text{Lip}(Y) \rightarrow D^{1,q}(X)$ is bounded if and only if

1. φ belongs to the Reshetnyak class $D_{\text{loc}}^{1,q}(X; Y)$,
2. φ **has finite distortion**, i.e. $|D\varphi| = 0$ a.e. on the set $Z_\varphi := \{\mathcal{J}\varphi(x) = 0\}$,
3. the operator distortion function $K_{q,p}(\cdot, \varphi)$ belongs to $L^\varkappa(X)$, $1/\varkappa = 1/q - 1/p$.

Moreover, $\|\varphi^*\| = \|K_{q,p}(\cdot, \varphi)\|_{L^\varkappa(X)}$.

The proof of necessity for $q < p$: main ideas

Let $\varphi: X \rightarrow Y$ be a homeomorphism, and suppose that φ^* is bounded. Hence, for every $u \in D^{1,p}(Y) \cap \text{Lip}(Y)$, $u \circ \varphi \in D^{1,q}(X)$ and

$$\|u \circ \varphi\|_{D^{1,q}(X)} \leq \|\varphi^*\| \|u\|_{D^{1,p}(Y)}.$$

Question

How can we extract geometric properties of φ ?

A measure associated with the composition operator

We construct a finite regular Borel measure Ψ on Y such that the inequality

$$\int_{\varphi^{-1}(B)} |\nabla(\varphi^* u)|^q d\mu \leq \Psi(B)^{\frac{q}{\varkappa}} \left(\int_B |\nabla u|^p d\nu \right)^{\frac{q}{p}}$$

holds for every $B \in \mathcal{B}(Y)$ and every $u \in D^{1,p}(Y) \cap \text{Lip}(Y)$. Here $1/\varkappa = 1/q - 1/p$.

It is clear that $\Psi(Y) = \|\varphi^*\|^\varkappa$.

Measure Ψ “contains” **geometric properties** of the mapping φ .

Regularity of the homeomorphism: I

Fix a bounded open set $U \subset Y$. By the definition of Ψ , we have

$$\begin{aligned} \int_{\varphi^{-1}(U)} |\nabla(\varphi^* u)|^q d\mu &\leq \Psi(U)^{\frac{q}{\varkappa}} \left(\int_U |\nabla u|^p d\nu \right)^{\frac{q}{p}} \\ &\leq \Psi(U)^{\frac{q}{\varkappa}} \left(\int_U \text{Lip}(u; U)^p d\nu \right)^{\frac{q}{p}} = \Psi(U)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}} \text{Lip}(u; U)^q. \end{aligned}$$

Regularity of the homeomorphism: II

$$\text{(previous slide)} \hookrightarrow \int_{\varphi^{-1}(U)} |\nabla(\varphi^* u)|^q d\mu \leq \Psi(U)^{\frac{q}{p}} \nu(U)^{\frac{q}{p}} \text{Lip}(u; U)^q.$$

Hence, the operator

$$\varphi_U^*: \text{Lip}(U) \rightarrow D^{1,q}(\varphi^{-1}(U))$$

is bounded.

We conclude that $\varphi \in D^{1,q}(\varphi^{-1}(U); U)$, and we have the inequality

$$\int_{\varphi^{-1}(U)} |D\varphi|^q d\mu = \|\varphi^*\|^q \leq \Psi(U)^{\frac{q}{p}} \nu(U)^{\frac{q}{p}}.$$

In particular, since φ is a homeomorphism we have $\varphi \in D_{\text{loc}}^{1,q}(X; Y)$.

The homeomorphism has finite distortion

$$\text{(previous slide)} \hookrightarrow \int_{\varphi^{-1}(U)} |D\varphi|^q d\mu \leq \Psi(U)^{\frac{q}{s}} \nu(U)^{\frac{q}{p}}$$

Let $Z_\varphi = \{x \in X \mid \mathcal{J}\varphi(x) = 0\}$. Then $\nu(\varphi(Z_\varphi)) = 0$, and hence

$$\int_{Z_\varphi} |D\varphi|^q d\mu \leq \Psi(U)^{\frac{q}{s}} \nu(U)^{\frac{q}{p}} = 0.$$

It follows that $|D\varphi| = 0$ a.e. on Z_φ .

From this, we can prove that Ψ is absolutely continuous with respect to ν . We denote the Radon–Nykodym density by Ψ' .

Integrability of the operator distortion function: I

$$\text{(previous slide)} \hookrightarrow \int_{\varphi^{-1}(U)} |D\varphi|^q d\mu \leq \left(\int_U \Psi' d\nu \right)^{\frac{q}{\varkappa}} \nu(U)^{\frac{q}{p}}$$

For simplicity, suppose that φ has the Lusin N -property. Let $B \in \mathcal{B}(X)$ be a Borel subset, and put $U = \varphi(B)$. By the change-of-variable, we obtain

$$\begin{aligned} \int_B |D\varphi|^q d\mu &\leq \left(\int_{\varphi(B)} \Psi' d\nu \right)^{\frac{q}{\varkappa}} \left(\int_B \mathcal{J}\varphi(x) d\mu \right)^{\frac{q}{p}} \\ &= \left(\int_B \Psi'(\varphi(x)) \mathcal{J}\varphi(x) d\mu \right)^{\frac{q}{\varkappa}} \left(\int_B \mathcal{J}\varphi(x) d\mu \right)^{\frac{q}{p}}. \end{aligned}$$

A converse to Hölder's inequality: I

Let $1/p + 1/p' = 1$, $1 < p < \infty$.

Suppose that f , g , h are nonnegative measurable functions. If $f(x) \leq g(x)h(x)$ a.e., then

$$\int_B f \, d\mu \leq \left(\int_B g^p \, d\mu \right)^{\frac{1}{p}} \left(\int_B h^{p'} \, d\mu \right)^{\frac{1}{p'}}$$

for every measurable B (Hölder's inequality).

A converse to Hölder's inequality: II

Let $1/p + 1/p' = 1$, $1 < p < \infty$.

Theorem (A converse to Hölder's inequality)

Suppose that f , g , h are nonnegative measurable functions such that the inequality

$$\int_B f \, d\mu \leq \left(\int_B g^p \, d\mu \right)^{\frac{1}{p}} \left(\int_B h^{p'} \, d\mu \right)^{\frac{1}{p'}}$$

holds for every measurable B . Then $f(x) \leq g(x)h(x)$ a.e.

Integrability of the operator distortion function: II

(previous slide) $\hookrightarrow$
$$\int_B |D\varphi|^q d\mu \leq \left(\int_B \Psi'(\varphi(x)) \mathcal{J}\varphi(x) d\mu \right)^{\frac{q}{\kappa}} \left(\int_B \mathcal{J}\varphi(x) d\mu \right)^{\frac{q}{p}}$$

From the converse to Hölder's inequality ($q/\kappa + q/p = 1$) we obtain the inequality

$$|D\varphi|^q(x) \leq (\Psi'(\varphi(x)) \mathcal{J}\varphi(x))^{\frac{q}{\kappa}} \mathcal{J}\varphi(x)^{\frac{q}{p}}$$

Hence, the operator distortion function has an L^κ -majorant:

$$K_{q,p}(x, \varphi) := \frac{|D\varphi|(x)}{\mathcal{J}\varphi(x)^{\frac{1}{p}}} \leq (\Psi'(\varphi(x)) \mathcal{J}\varphi(x))^{\frac{1}{\kappa}},$$

if $\mathcal{J}\varphi(x) \neq 0$. We set $K_{q,p}(x, \varphi) := 0$ if $\mathcal{J}\varphi(x) = 0$.

Integrability of the operator distortion function: III

$$\text{(previous slide)} \hookrightarrow K_{q,p}(x, \varphi) \leq (\Psi'(\varphi(x))\mathcal{J}\varphi(x))^{\frac{1}{\varkappa}}$$

By integrating $K_{q,p}(\cdot, \varphi)^{\varkappa}$ over X , we obtain the inequality

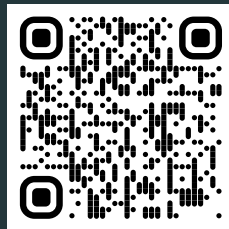
$$\int_X K_{q,p}(x, \varphi)^{\varkappa} d\mu(x) \leq \int_X \Psi'(\varphi(x))\mathcal{J}\varphi(x) d\mu(x) = \int_Y \Psi'(y) d\nu(y) = \Psi(Y) = \|\varphi^*\|^{\varkappa}.$$

The inequality $\|\varphi^*\| \leq \|K_{q,p}(\cdot, \varphi)\|_{L^{\varkappa}(X)}$ follows from the proof of sufficiency.

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Q&A



Slides

Brief history: I

- In 1941, S. L. Sobolev proved that C^1 quasi-isometries preserve the $L^{1,p}$ spaces in $\mathbb{R}^n$, $n \geq 2$.

He conjectured that all isomorphisms of Sobolev spaces $L^{1,p}$ are quasi-isometries.

- It follows from V. G. Maz'ya PhD thesis that C^1 isomorphisms of $L^{1,p}$ spaces are
 - quasi-isometries if $p \neq n$,
 - quasiconformal mappings if $p = n$.

Brief history: II

In 1968, Yu. G. Reshetnyak posed the problem of characterizing all isomorphisms of the Sobolev space $L^{1,n}$ induced by quasiconformal mappings.

- In 1974, E. A. Poletsky answered this question for the space $L^{1,n} \cap C^1$.
- In 1975, S. K. Vodopyanov and V. M. Gol'dshtejn gave a full solution of Reshetnyak's problem.

In 1980s, the study of mappings that induce bounded composition operators on Sobolev spaces began.

Brief history: III

Characterization of homeomorphisms that induce bounded composition operators on Sobolev spaces were established in model spaces:

- on Carnot groups (also in Euclidean space), see S. K. Vodopyanov and A. D. Ukhlov (1998), S. K. Vodopyanov and N. A. Evseev (2022);
- on Riemannian manifolds, see S. K. Vodopyanov (2024 and 2025).